

Question			Marking details	Marks available				Maths	Prac
				AO1	AO2	AO3	Total		
2	(a)	(i)	$\rho = \frac{RA}{L}$ correct transposition at any stage or by implication (1) $A = \pi \left(\frac{d}{2}\right)^2$ or equivalent correctly used to give value in m ² or by implication (1) $\rho = 4.9 \times 10^{-7} \Omega \text{ m}$ [accept $4.90 \times 10^{-7} \Omega \text{ m}$] UNIT (1) Only one mark lost for $20 \times 10^{-7} \Omega \text{ m}$ or $19.6 \times 10^{-7} \Omega \text{ m}$ No sig fig penalty	1	1		3	3	3
		(ii)	One mark penalty for working with uncertainties of half the resolutions without comment. No penalty if reasonable justification given. This applies to both methods below. Fractional or percentage uncertainties method In r : 0.02[1] or 2[.1]%. In R : 0.04 5 or 4.5% In L : 0 or 0.1% (1) Correct composition of individual fractional or percentage uncertainties (0.087 (8.7%) and 0.09 (9%)) (1) So 0.4 or $0.5 \times 10^{-7} [\Omega \text{ m}]$ if $4.9 \times 10^{-7} [\Omega \text{ m}]$ given in (i) or 0.39 to $0.44 \times 10^{-7} [\Omega \text{ m}]$ if $4.90 \times 10^{-7} [\Omega \text{ m}]$ given in (i) ecf (1) <i>Only one mark penalty for omitting 10^{-7} in this part.</i> Maximum value method $\frac{2.3 \times \pi \times (0.245 \times 10^{-3})^2}{0.811}$ or by implication (1) 5.34 or $5.35 \times 10^{-7} [\Omega \text{ m}]$, or 5.3 or $5.4 \times 10^{-7} [\Omega \text{ m}]$ (1) So ± 0.4 or $0.5 \times 10^{-7} [\Omega \text{ m}]$ if $4.9 \times 10^{-7} [\Omega \text{ m}]$ given in (i) or 0.43 , 0.44 or $0.45 \times 10^{-7} [\Omega \text{ m}]$ if $4.90 \times 10^{-7} [\Omega \text{ m}]$ in (i) (1) Answer to 1 or 2 sig figs only		3		3	3	3

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	(b)		Longer wire (1) Don't accept more repeat readings Less percentage or fractional uncertainty in resistance (1) Don't accept less percentage or fractional uncertainty in length Or winding wire or equivalent to measure across several diameters at once (1) Less percentage error in diameter (1)			2	2		2
			Question 2 total	1	5	2	8	6	8